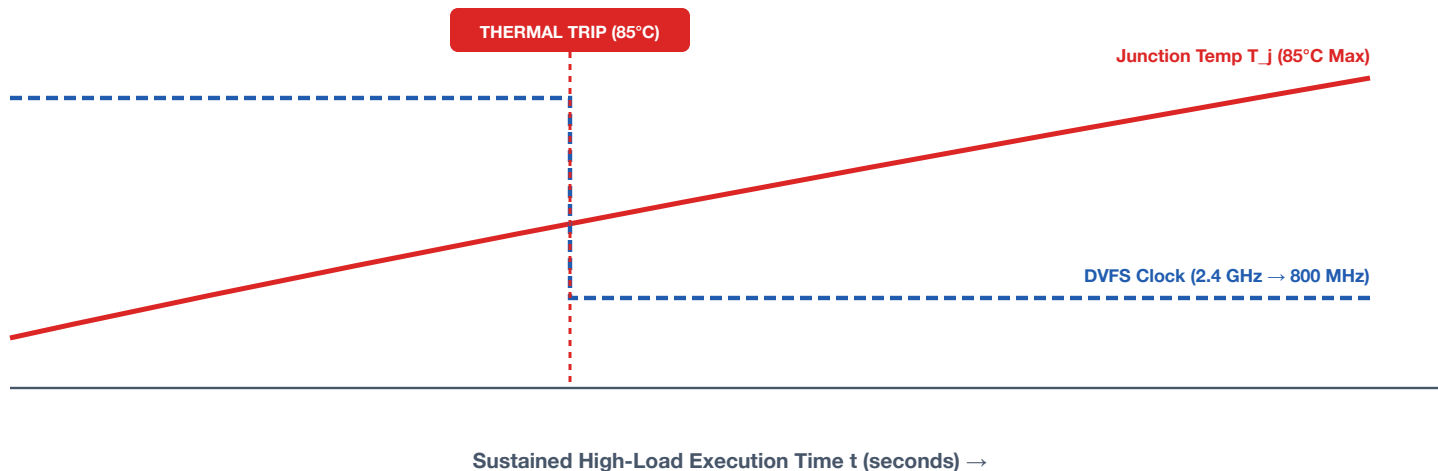


THERMODYNAMIC JUNCTION HEATING & DVFS DOWNCLOCKING RIPPLE

Silicon Thermal RC Dynamics: $T_j(t) = T_{\text{ambient}} + P \cdot \theta_{JA} \cdot (1 - \exp(-t / \tau_{th}))$



THE THERMAL DERATING CASCADING FAILURE

1. NPU thermal throttle cuts frequency by 3× ⇒ Inference latency triples from 20 ms to 60 ms.
2. Stopping distance d_{reaction} triples from 10 cm to 30 cm ⇒ Robot breaches clearance buffer unless enforcer backs off speed.